SignalEdge

AI-Powered Fraud Detection & Identity Verification Platform

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Plan

Problem Statement

Despite advancements, financial institutions face significant challenges in real-time fraud detection and identity verification.

Real-time Detection Gap
Financial institutions lack effective real-time fraud detection, leading to delayed responses.

Manual Process Limitations
Manual reviews lead to errors and are not scalable with increasing transaction volumes.

Inconsistent Verification
There's a lack of consistency in verifying identities, leading to potential serious risks.

Proposed Solution

Our platform offers an integrated, AI-driven solution that seamlessly combines real-time fraud detection with advanced identity verification

Real-Time Fraud Detection
Utilize AI-driven risk scoring to monitor and flag suspicious transactions instantaneously.

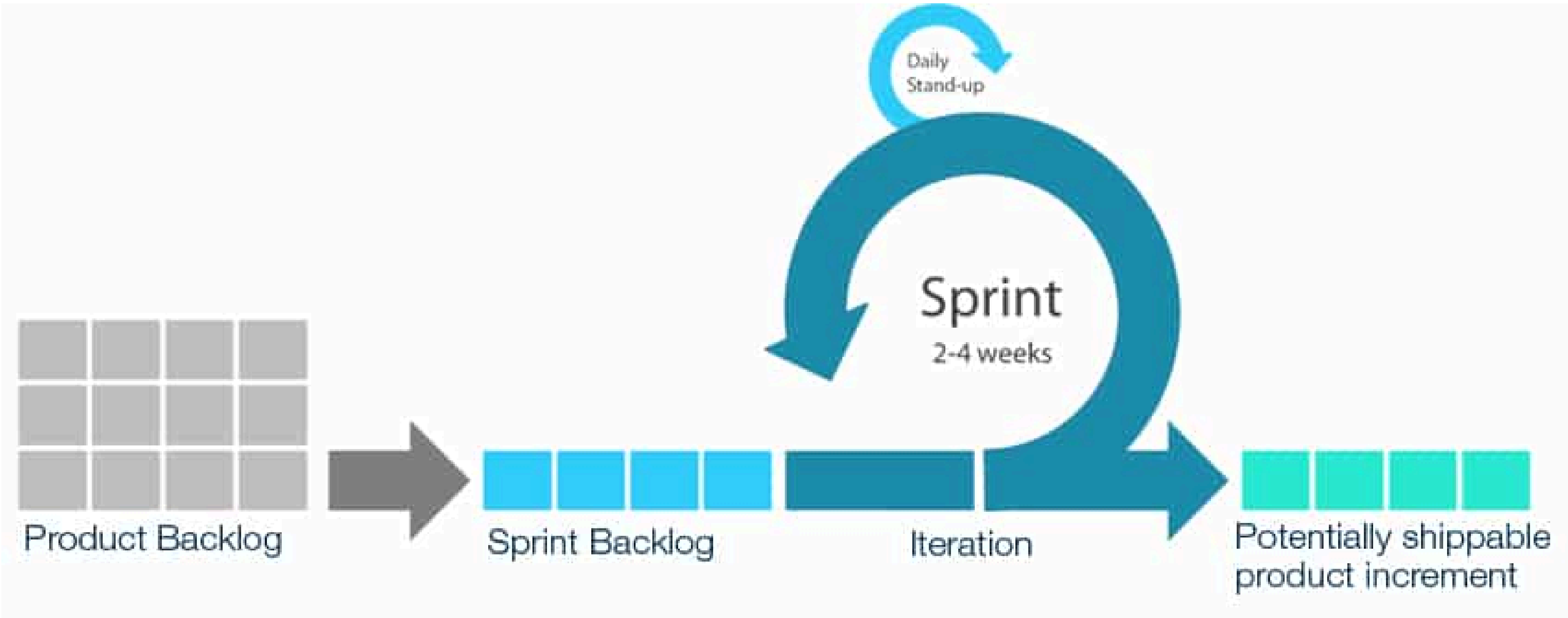
AI-Powered Identity Verification
Detect and prevent the use of AI-generated fake identity documents during KYC processes.

Streamlined Review Workflow
Provide analysts with intuitive tools for efficient review, approval, and rejection of flagged items.

Actionable Analytics
Deliver insights through geographic heatmaps, trend analysis, and merchant risk evaluations.

AI Model Management
Enable continuous monitoring, comparison, and optimization of fraud detection models.

Methodology & Technical Choices



Scrum, an Agile methodology, empowers the team to adapt quickly to changes, deliver incremental value, and continuously refine the product for optimal outcomes

Fast & on point Delivery

Regular Team communication

Coping with change

Consistent growth

Scrum Team

Ahmed Khelifi

Product Owner

Ahmed Khelifi

Software Engineer

TEK-UP

Scrum Master

Technical Choices

Software Development

REST API
Web Socket using Daphne
Test Unitaire - pytest
Postgresql

Headless UI component
Axios for Http Requests
React Testing Library + Jest

Hybrid Mobile App
EAS for CI/CD Workflow

Technical Choices

Deep Learning & Machine Learning

Fine-Tunning Resnet 50
Exporting Modal ONNX
Used For Deep Fake Detection

For Transactions Detection

Manage Datasets
Model dev & Training

Technical Choices

Infra & Devops

create and manage containers

orchestration

Deployment

Host & Deploy AI Models

Planification

Actors Identification

Super Admin

Total Visibility

Analyst

Monitor Transactions & KYC Submissions

AI Engineer

Manage Models & Datasets

Finance Consultant

Access Transaction Data

Non-Functional Requirements

Performance
The platform should handle a high volume of transactions with minimal latency.

Scalability
It should scale to accommodate growing data and users

Security
Protect sensitive data and ensure compliance with international standards

Usability
The interface should be intuitive and user-friendly.

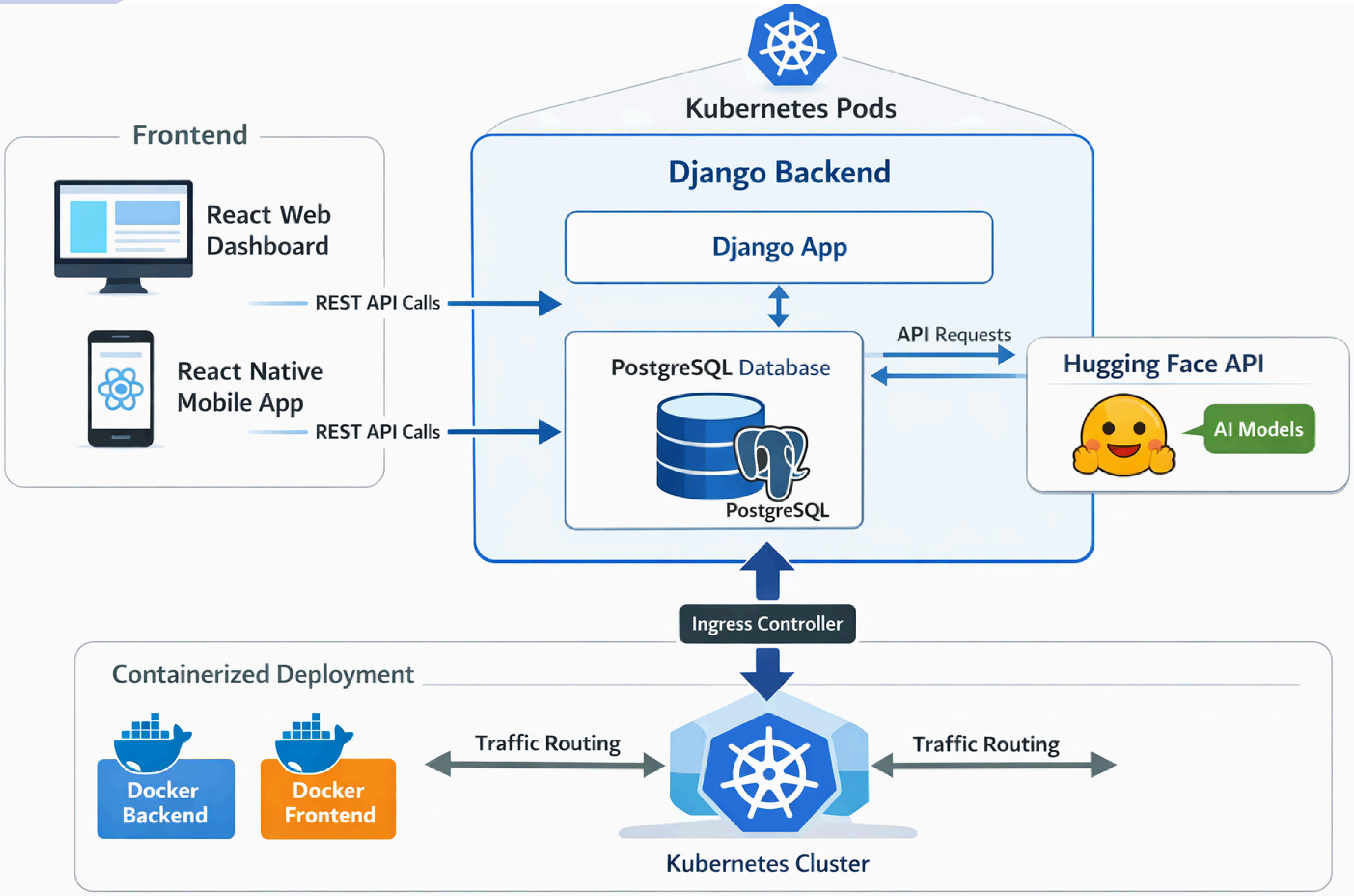
Maintainability
Code should be modular and easy to update

Functional Requirements

Authentication	Real-Time Fraud Detection	KYC Verification	Alert Management
Secure login and user management.	Monitor transactions and flag suspicious activity instantly.	Validate identity documents and detect AI-generated fakes.	Notify analysts of flagged transactions and manage alerts.

Analytics and Reporting	Model Management
Provide dashboards with insights, heatmaps, and trend analysis.	Allow for monitoring, updating, and comparing AI models.

System Architecture



Kubernetes Pods
Frontend
React Web Dashboard
REST API Calls
React Native Mobile App
REST API Calls
Django Backend
Django App
PostgreSQL Database
PostgreSQL
API Requests
Hugging Face API
AI Models
Ingress Controller
Containerized Deployment
Docker Backend
Docker Frontend
Traffic Routing
Traffic Routing
Kubernetes Cluster

Demo

Check out the full demo on

YouTube

[*Click here*](https://youtu.be/YYAs1BpowNs)

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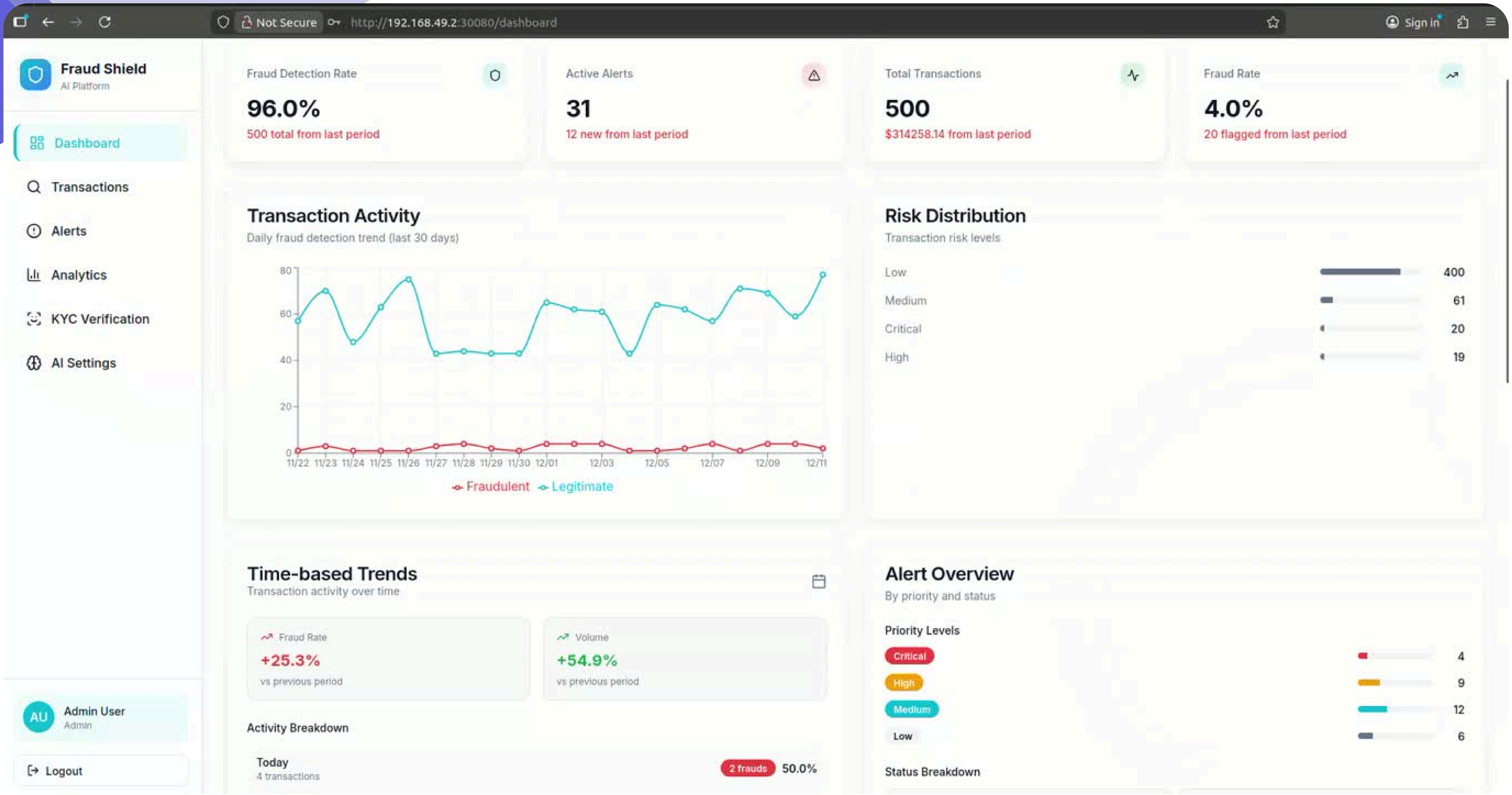
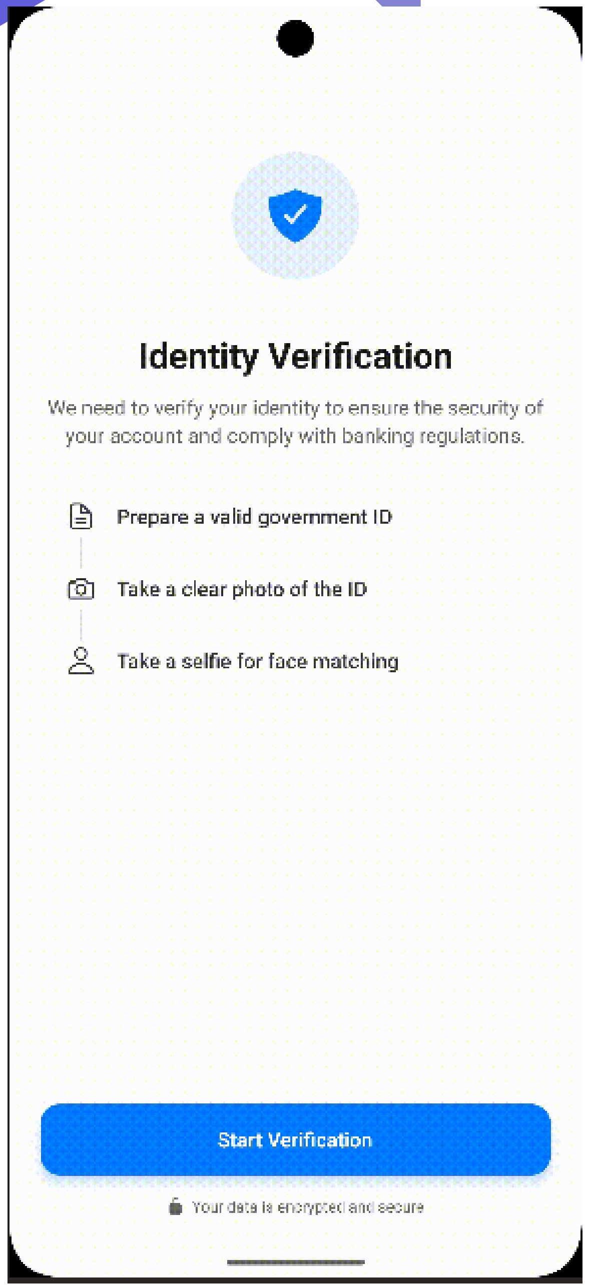
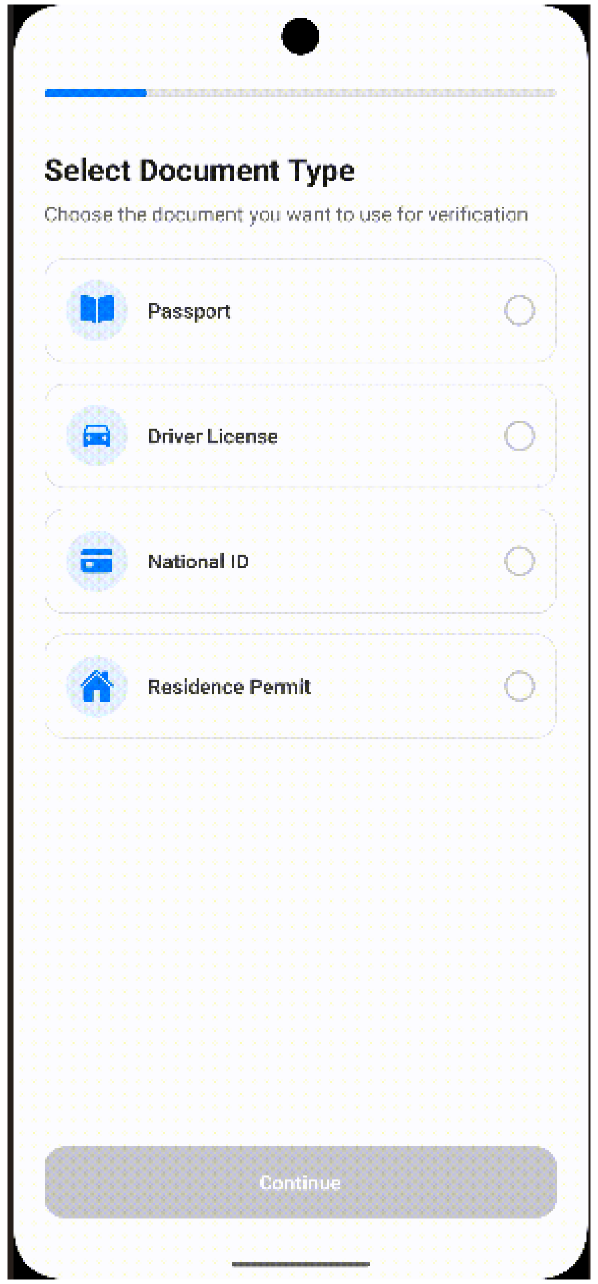
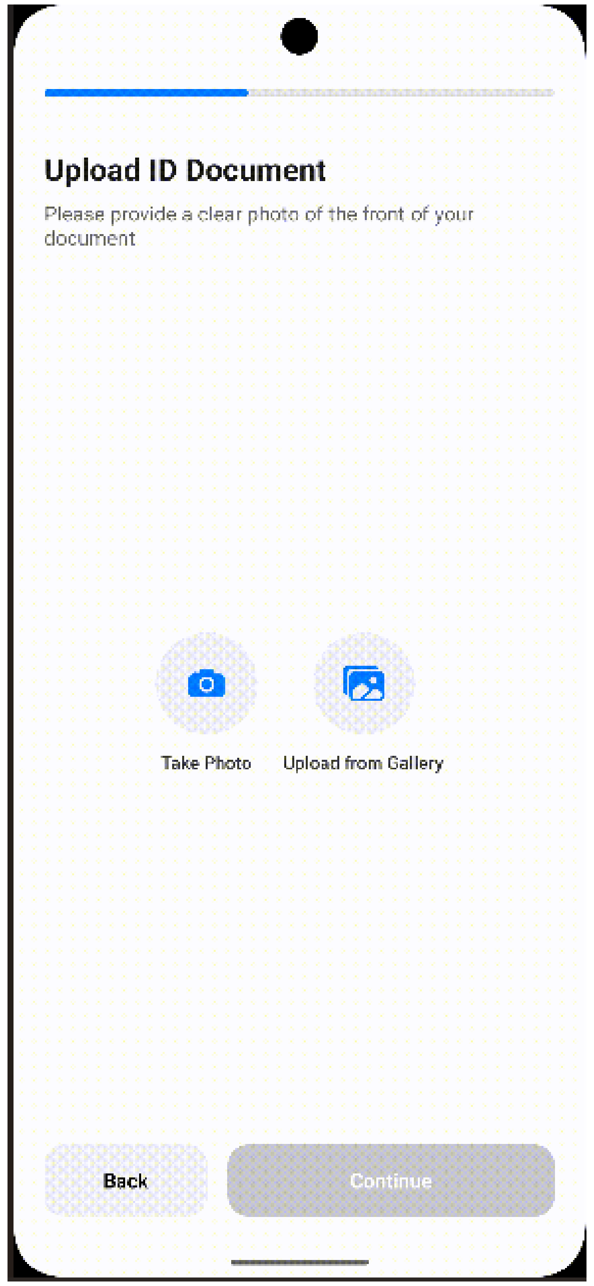
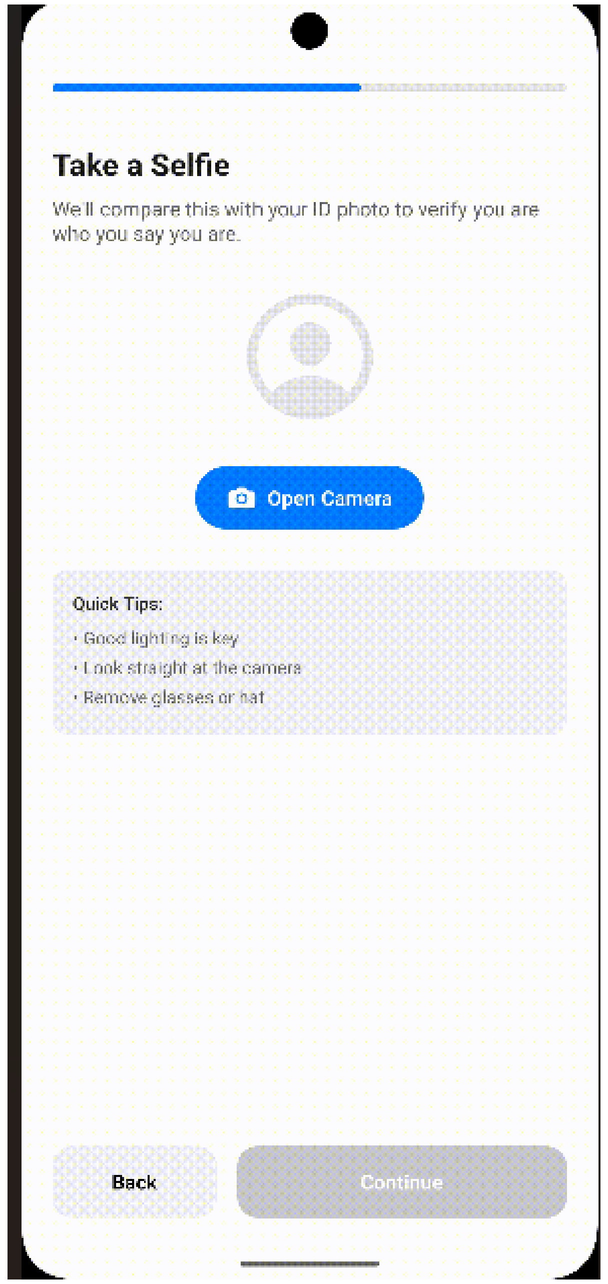

Not Secure
http://192.168.49.2:30080/dashboard
Sign in
Fraud Shield
AI Platform
Dashboard
Transactions
Alerts
Analytics
KYC Verification
AI Settings
AU
Admin User
Admin
Logout
Fraud Detection Rate
96.0%
500 total from last period
Active Alerts
31
12 new from last period
Total Transactions
500
$314258.14 from last period
Fraud Rate
4.0%
20 flagged from last period
Transaction Activity
Daily fraud detection trend (last 30 days)
80
60
40
20
0
11/22 11/23 11/24 11/25 11/26 11/27 11/28 11/29 11/30 12/01 12/03 12/05 12/07 12/09 12/11
Fraudulent
Legitimate
Risk Distribution
Transaction risk levels
Low 400
Medium 61
Critical 20
High 19
Time-based Trends
Transaction activity over time
Fraud Rate
+25.3%
vs previous period
Volume
+54.9%
vs previous period
Activity Breakdown
Today
4 transactions
2 frauds
50.0%
Alert Overview
By priority and status
Priority Levels
Critical 4
High 9
Medium 12
Low 6
Status Breakdown

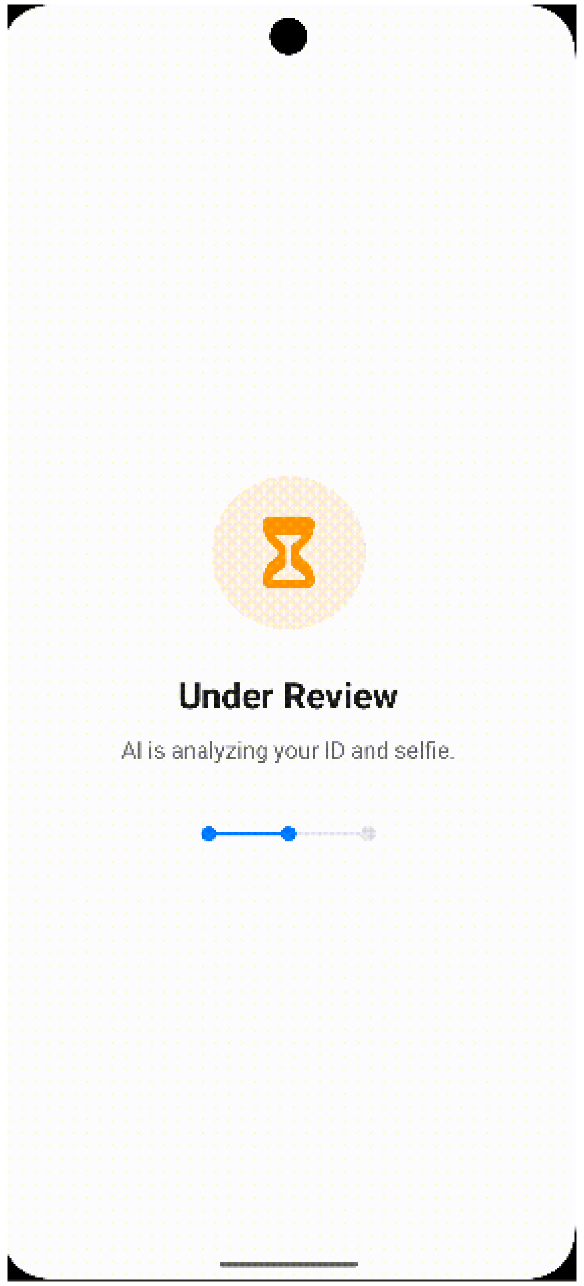
React Native - Mobile App











Thank You